

Computer Networks: Unit 1 Network Layer and Routing

VTU 2022 Scheme CBCS - BCS603 Module 1

--- MODULE OVERVIEW & LEARNING OBJECTIVES ---

1. Network Layer Design: Store-and-Forward Packet Switching vs Circuit Switching
2. IPv4 Header format, Classless Inter-Domain Routing [CIDR], and Subnetting math
3. Address Resolution Protocol [ARP] and Dynamic Host Configuration Protocol [DHCP]
4. Routing Algorithms: Distance Vector Routing [Bellman-Ford] and Count-to-Infinity problem
5. Link State Routing: Dijkstra Algorithm and Open Shortest Path First [OSPF]

--- IMPORTANT VTU 10-MARK QUESTIONS ---

- Q1. Subnet the IP 192.168.1.0/24 into 4 equal subnets with host ranges [10 Marks]
- Q2. Explain Bellman-Ford routing algorithm with 5-node graph [10 Marks]